

HOMESCOPE

Assignment 3 - Project Prototype Documentation

AI-Powered American House Price Dashboard

Prototype documentation for a housing market dashboard that helps users explore market averages, compare a listing against similar homes, and review a model-driven fair value prediction in a clear visual interface.

Student	Course	Assignment	Prototype Tool
Eric Reyes	Project & Portfolio IV	Assignment 3 - Project Prototype	Figma

Prototype Link

Playable prototype: [Open HomeScope Figma Prototype](https://www.figma.com/proto/TlkjH4ZRzmOaYOxFLpnTy/HomeScope-Prototype?node-id=7-2&p=f&viewport=88%2C472%2C0.22&t=Vqz55rHM3794e9dn-8&scaling=min-zoom&content-scaling=fixed&starting-point-node-id=7%3A2&page-id=0%3A1&hide-ui=1)

Full URL: <https://www.figma.com/proto/TlkjH4ZRzmOaYOxFLpnTy/HomeScope-Prototype?node-id=7-2&p=f&viewport=88%2C472%2C0.22&t=Vqz55rHM3794e9dn-8&scaling=min-zoom&content-scaling=fixed&starting-point-node-id=7%3A2&page-id=0%3A1&hide-ui=1>

Project and Prototype Summary

HomeScope is an AI-powered American house price dashboard focused on making housing data easier to understand. The prototype represents the planned user experience before final development by showing how a user moves from market exploration, to model evaluation, to a fair value prediction workflow.

The prototype is desktop-first and uses a dark dashboard interface with cards, filters, chart panels, model evidence, and plain-language result labels. This helps me plan the interface before moving deeper into development.

Similar Products Researched

Product	Smart UX Decisions Observed	What HomeScope Borrows
Zillow	Strong property search, listing cards, neighborhood context, and consumer-friendly value estimates.	HomeScope borrows familiar search/filter flow and simple property comparison language.
Redfin	Clean listing experience, home estimates, and market dashboard-style context.	HomeScope borrows dashboard-style market metrics and readable comparison panels.
Realtor.com	MLS-style property discovery, property details, and local market resources.	HomeScope borrows location-based browsing and detailed listing comparison structure.

Where HomeScope Improves the UX

- Transparent model evidence: HomeScope shows Linear Regression vs. Random Forest instead of presenting a prediction with no context.
- Plain-language market label: the result screen translates model output into labels such as Above Average instead of forcing users to interpret raw model metrics.
- Market snapshot first: the dashboard shows matching listings, average price, median price, and price per square foot before asking the user to trust a prediction.
- Prediction assumptions are reduced: the prediction form keeps user input simple while the final app can fill hidden demographic or location features from selected market data.
- Data-focused differentiator: HomeScope is designed as a transparent research dashboard, not a live listing marketplace or brokerage funnel.

Prioritized GUI Feature List

#	Feature	Priority	User Need Met	Prototype Status
1	Dashboard navigation	Primary	Lets users move between overview, market dashboard, model evidence, prediction form, and result screen.	Included
2	Market filters	Primary	Lets users narrow market context by state, city, beds, baths, and living space.	Included
3	Market metric cards	Primary	Shows matching listings, average price, median price, and average price per square foot.	Included
4	Price distribution chart	Primary	Helps users see how prices are spread in the selected market.	Included
5	Price vs. living space chart	Primary	Shows the relationship between home size and price.	Included
6	Average price by city chart	Secondary	Gives local comparison context across cities.	Included
7	National trend context chart	Secondary	Adds broader U.S. price trend context.	Included
8	Model evaluation screen	Primary	Compares Linear Regression and Random Forest using MAE, RMSE, and R-squared.	Included
9	Prediction form	Primary	Allows the user to enter sample listing details.	Included
10	Prediction result screen	Primary	Displays predicted fair value, listing price, market label, and explanation.	Included

Current Prototype Capabilities

The current Figma prototype supports the planned screen flow and the major user interactions needed for Assignment 3. It does not run the real machine learning model, but it accurately represents how the final interface will organize filters, charts, model comparison, prediction input, and result feedback.

- Landing Overview introduces HomeScope and provides entry points for market exploration or prediction.
- Market Dashboard shows filters, key market metrics, and chart panels for visual comparison.
- Model Evaluation shows the planned model comparison table and residual plot area.
- Prediction Form shows the user inputs needed for a sample listing.
- Prediction Result shows fair value, listing price, market label, comparison chart, and plain-language insight.

Prototype Interaction Flow

Starting Screen	User Action	Destination Screen
Landing Overview	Explore Market	Market Dashboard
Landing Overview	Compare Listing	Prediction Form
Landing Overview	Predict Fair Value	Prediction Form
Market Dashboard	View Model Evidence	Model Evaluation
Market Dashboard	Predict Fair Value	Prediction Form
Model Evaluation	Back to Dashboard	Market Dashboard
Model Evaluation	Open Prediction Form	Prediction Form
Prediction Form	Predict Fair Value	Prediction Result
Prediction Result	Back to Dashboard	Market Dashboard
Prediction Result	Edit Inputs	Prediction Form

Prototype Screens Included

Screen	Purpose
01 Landing Overview	Introduces the product and gives the user primary entry points.
02 Market Dashboard	Shows filters, market metrics, distribution chart, size/price chart, city comparison, and trend context.
03 Model Evaluation	Shows model comparison metrics and residual plot placeholder.
04 Prediction Form	Collects the sample property details used by the prediction workflow.
05 Prediction Result	Shows predicted fair value, listing price, market label, chart comparison, and explanation.
06 Documentation Notes for PDF	Used as an internal helper screen for documentation notes; the official documentation is this PDF.

UI/UX Considerations

- The interface is desktop-first because HomeScope is a data-heavy dashboard with multiple charts and comparison cards.
- Cards and accent colors separate different kinds of information: metrics, charts, model evidence, and action buttons.

- The prediction result uses plain language because regular users may not understand regression model output without translation.
- The visual design is consistent with the original HomeScope pitch: dark background, teal accent, rounded panels, and dashboard-style data visualization.
- The prototype avoids login, user settings, and account screens because they do not prove the unique HomeScope experience.

Planned Improvements After Feedback

- Improve chart labeling once real data outputs are finalized.
- Add more detailed tooltip explanations for MAE, RMSE, and R-squared.
- Consider adding an optional advanced section explaining hidden location and demographic assumptions used by the model.
- Refine the prediction result language so users understand it is a model estimate, not a guaranteed appraisal.

Conclusion

The HomeScope prototype demonstrates the complete planned interface flow for Assignment 3. It shows how users will explore market averages, review model evidence, submit a sample listing, and interpret a predicted fair value result. The prototype is ready for feedback and can guide the next development stage of the Project & Portfolio IV application.